AI AGENTS HACKATHON · AMERICAN UNIVERSITY IN DUBAI

Zill & Shams

ظل وشمس *“shade and sun”*

Shade when you need it. Sun when you want it.

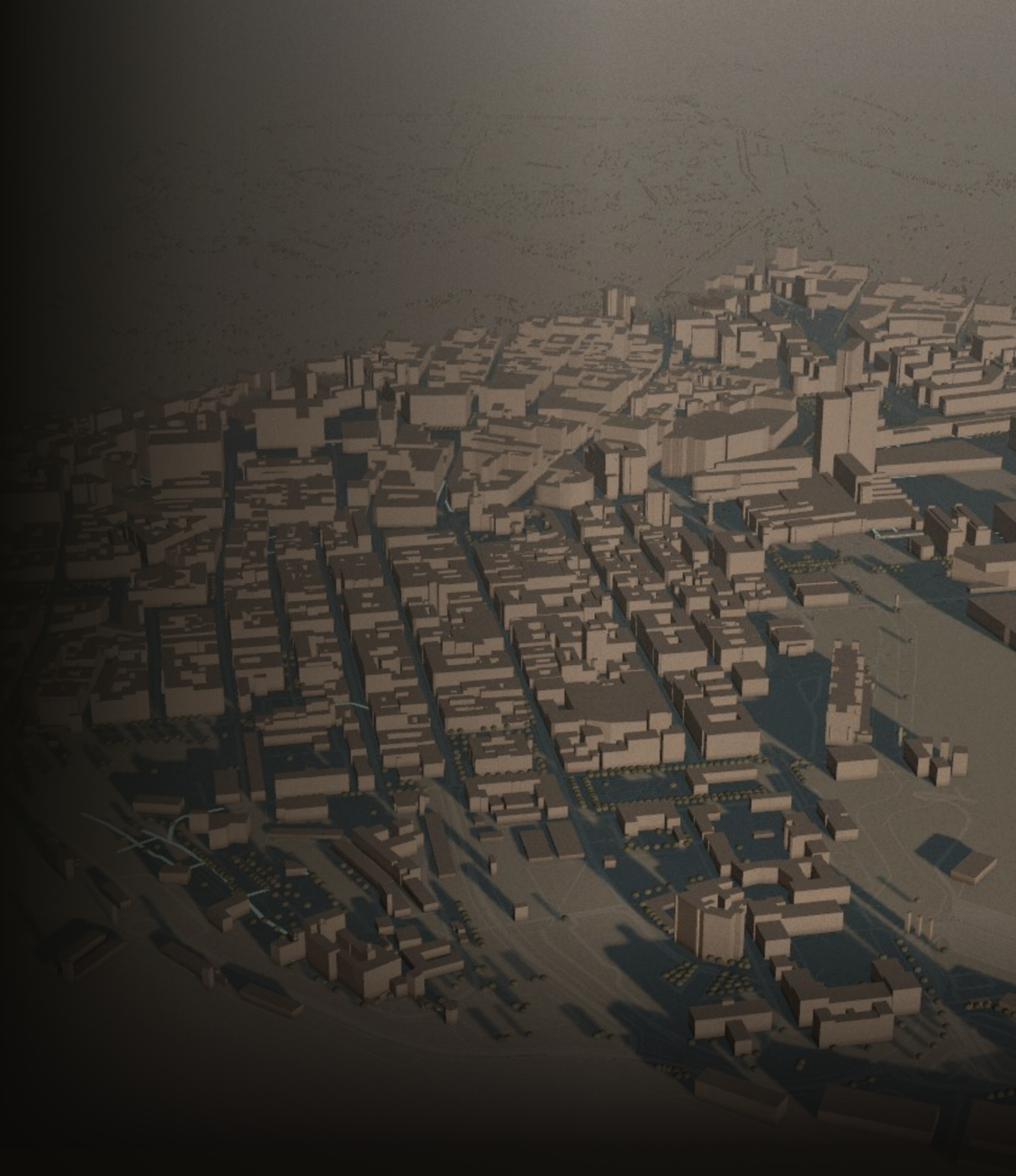
An AI agent that plans every walk and ride around the sun.

Maps plan around time. Nobody plans around the sun.

01 In hot cities, the route you pick decides how much direct sun you take, and in summer that is a health risk.

02 In dark winters, people get too little daylight. Oslo has about 6 hours of daylight in late December.

03 The sun and shade move every minute, so a static “shady street” or “sunny street” list is never right for long.



03 · WHY IT MATTERS

72%

of 12,346 adults in an Abu Dhabi study were vitamin D deficient, and another 10% insufficient, in one of the sunniest places on earth.

BMC RESEARCH NOTES, 2019

70,000+

delivery riders work across Dubai.

DUBAI RTA

18.72 million

international overnight visitors came to Dubai in 2024.

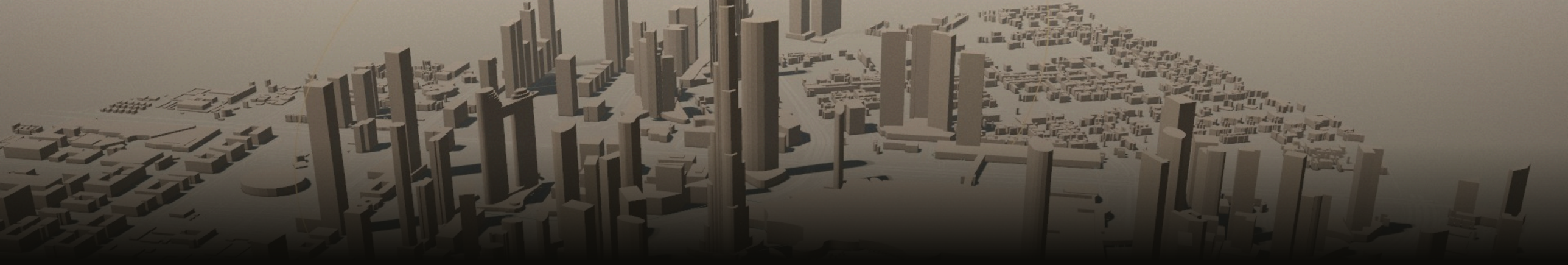
DUBAI DET VIA WAM

12:30–15:00

The UAE bans outdoor work in direct sun, 15 June to 15 September, for the 21st year in 2025.

UAE MINISTRY OF HUMAN RESOURCES AND EMIRATISATION

Sources: BMC Research Notes 2019, vitamin D in Abu Dhabi (n=12,346): ncbi.nlm.nih.gov/pmc/articles/PMC6693241 · Dubai RTA, delivery rider programmes: rta.ae (news: 23 temporary rest areas to protect delivery riders from heat stress) · Dubai DET via WAM: wam.ae/article/bi46kc8-dubai-welcomes-1872-million-international-visitors · MoHRE, Midday Break 2025: mohre.gov.ae/en/media-center/news/3/6/2025 · Oslo daylight at the December solstice (about 5 h 54 min): fjordtours.com/en/norway/people-and-culture/facts/winter-solstice



04 · THE PRODUCT

One engine, two modes.

SHADE MODE · FOR HOT CITIES

Finds the coolest route: the most shade and the least direct sun, using air-conditioned links such as the Dubai Metro to Dubai Mall walkway.

SUN MODE · FOR DAYLIGHT AND WELL-BEING

Finds the sunniest route for people who want more daylight: winter cities, office workers, anyone short on sun.

Same engine. The preference flips.

How it works, in one picture.

01 Open map data

→

02 3D model of the city

→

03 Where the sun and shadows fall, every 10 minutes

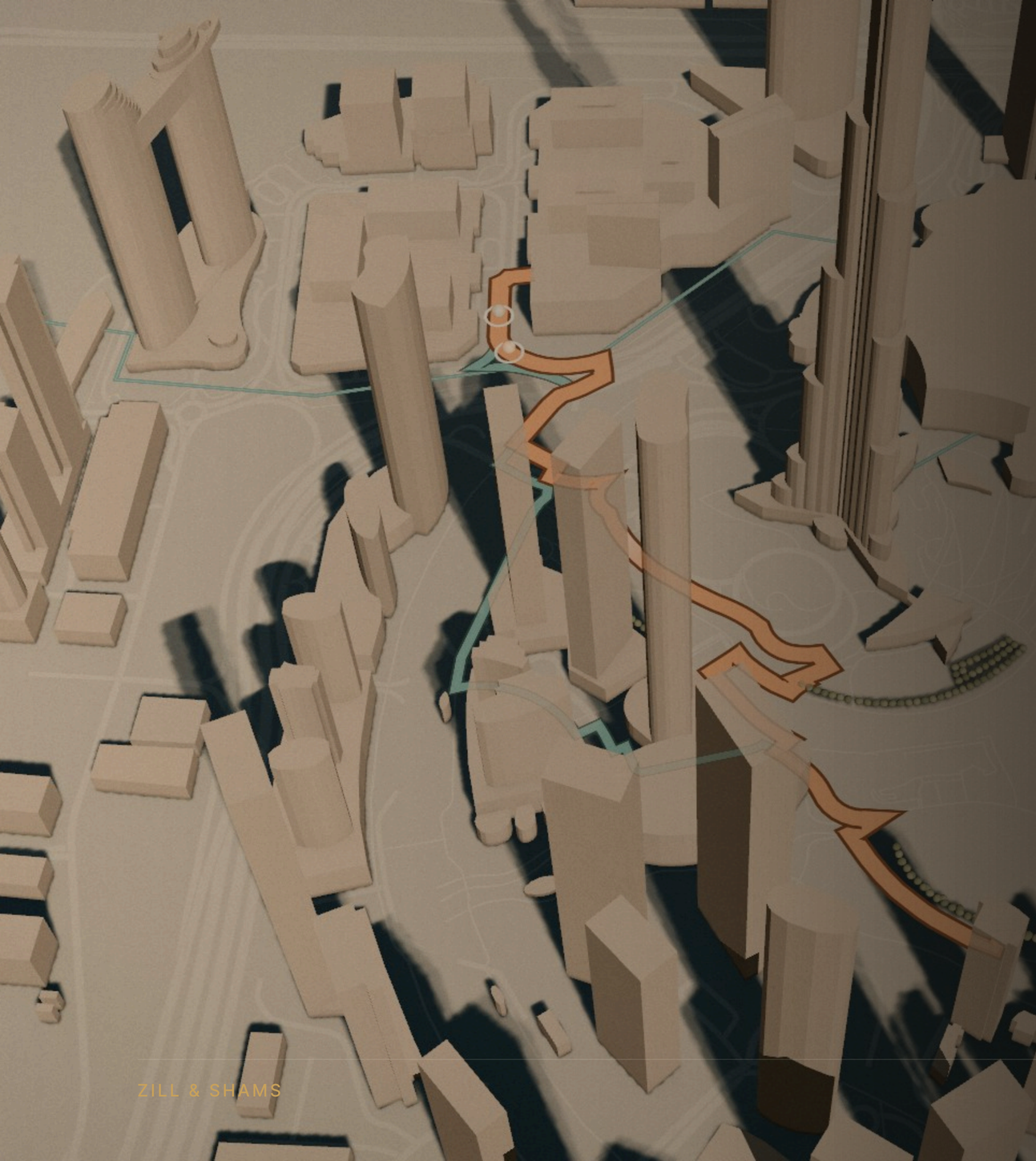
→

04 The best route for you

→

05 An AI agent that plans, checks and adjusts

Works for any city with OpenStreetMap data. Two cities live: Downtown Dubai (8,602 buildings, 7,659 street segments, 744 trees) and central Oslo.



06 · PROOF, SHADE MODE

A real walk computed by the engine: Burj Park → Business Bay Metro, a summer afternoon at 15:30.

6.2	40%	2.2
MINUTES LESS DIRECT SUN	LESS SUN	ONLY MINUTES LONGER

Computed from real map data, not a mock-up. Live at zill.itshamza.me

Model estimate; shade from buildings and mapped trees. Awnings and wind are not modelled yet.

Sun mode finds the sunnier walk: real routes, clear-sky geometry.

DUBAI · 15 APRIL · 10:30

Business Bay Metro → Burj Park

+8.7	+48%	+4.5
MIN MORE SUN	SUN VS FASTEST	MIN LONGER

Fastest: 1,101 s in direct sun of 1,435 s. Sunniest: 1,625 s of 1,704 s. Daylight window open: sun 59.9°, UV 8.7.

OSLO · 21 JUNE · 12:00

National Museum → Byporten Shopping

+11.3	+110%	+2.2
MIN MORE SUN	SUN VS FASTEST	MIN LONGER

Fastest: 613 s in direct sun of 1,259 s. Sunniest: 1,288 s of 1,390 s. Daylight window open: sun 51.0°, UV 6.8.

Routes chosen by a documented search over place pairs 400 to 1,500 m apart (data/scenarios/hero.json). Gains are model estimates from the engine, not field measurements.

08 · SECOND CITY

Oslo, built with the same pipeline.

No city-specific code: open map data in, a shade-aware city out.

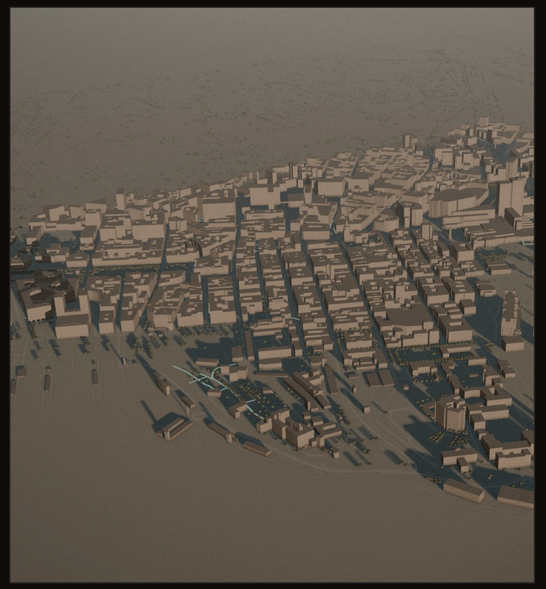
13,841
BUILDINGS IN THE SHADOW RING

10,082
STREET SEGMENTS

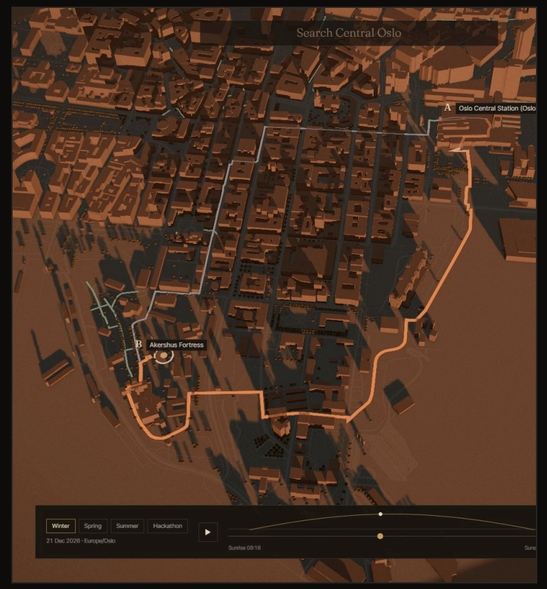
20,208
TREES

53
PLACES OF INTEREST

Low winter sun: at the December solstice the sun is about 6.6° high, so shadows run about 8.6 times a building’s height. The shadow engine is tested at exactly this angle.



FILM STILL · OSLO, LOW WINTER SUN



APP · OSLO, SUN MODE

Central Oslo, rendered from the baked city: the sun-mode route in the app, and the long shadows of a winter day in the film.

HONEST NOTE

Oslo’s winter weather is a labelled stand-in (the 2025 solstice). Oslo gets about 5 h 54 min of daylight at the solstice.

Source: data/cities/oslo/meta.json (counts); docs/PROGRESS.md; daylight length: fjordtours.com/en/norway/people-and-culture/facts/winter-solstice. Map data © OpenStreetMap contributors.

Daylight and heat, made visible.

UV INDEX

Today’s index with its WHO band, from low to extreme, and the matching sun-protection advice.

HEAT STRESS

A band from apparent temperature, labelled indicative.

DAYLIGHT WINDOW

Open when the sun is high (your shadow is shorter than you), UV is 3 or more and cloud is under 70%.

BREAK PROMPTS

Hydration and shade-break reminders on long, hot routes.

NOT MEDICAL ADVICE

Well-being and daylight guidance only. It does not diagnose or treat anything. Clinicians are future partners, not a claim we make today.

OSLO, IN WINTER

The sun is about 6° high and UV is near zero (0.1). Sun mode there is about daylight and mood, not vitamin D.



THE WELL-BEING CARD IN THE APP (DUBAI, 15 JAN, RECORDED WEATHER)

Oslo solstice scenario from docs/PROGRESS.md: sun 6.3°, shadow 9.1× height, UV 0.1, window closed. UV bands follow WHO categories.

An agent, not just a map.

PLANS Understands “I’m walking to the metro, I tire easily in the heat” and compares options.

CHECKS A second AI step verifies every claim; every number comes from the engine, never made up.

ADAPTS If you are delayed or a street is closed, it replans and asks before changing your plan.

A pluggable provider chain (OpenCode Go, then Voke) with automatic fallback. Until a live key is set, sessions are labelled recorded replays.



How accurate, how fresh.

INPUT	REFRESH
Sun position	Exact for any minute; within 0.5° of a reference solar model on 20 test samples
Shadows	Every 10 minutes, on a 2 m grid
Route timing	Departure checked every 5 minutes
Weather	Hourly: UV, cloud, sunlight, temperature
Map data	Re-baked from OpenStreetMap on demand; a weekly job is planned, not yet running

TESTED

Shadow length and direction for boxes at several sun angles, including Oslo's 6.6° winter sun. 470 automated tests pass (375 backend, 95 frontend).

LIMITS, SAID PLAINLY

About 79% of Dubai buildings have no mapped height, so typical heights are used. Trees and awnings are only what is mapped. A passing cloud is not predicted. No field validation yet, so we claim no accuracy percentage.

Source: docs/ACCURACY.md, docs/VALIDATION.md, docs/PROGRESS.md. Next step: walk the hero routes with a light sensor and compare.

Nothing to steal: we store no user data.

BY DESIGN

- No accounts, no user database
- Location used per request, then discarded
- Logs scrub coordinates and keys
- No ads, no third-party trackers

ON THE LIVE SITE

- Per-IP rate limits, tightest on the AI agent
- 64 KB request cap, 1,000-character text cap
- CORS locked to our domain
- CSP, HSTS, no framing (except the hotel widget)

SECRETS AND AUDIT

- LLM key stays on the server
- pip-audit: no known vulnerabilities
- npm audit: 2 findings, both in Next’s build-time PostCSS; the fix is a major upgrade

We aim to align with the UAE PDPL and the GDPR by design. This is an aim, not a certification. No external penetration test yet; it comes before any B2B launch.

Source: docs/SECURITY.md. Controls implemented in backend/zill/api/security.py and frontend/next.config.ts (2026-10-23).

Who pays.

SEGMENT	WHAT THEY GET	HOW THEY PAY
Delivery fleets targets such as Talabat, Deliveroo, Careem	Heat-aware routes and shift pacing for riders	Per rider per month pricing hypothesis: ~$1
Hotels and tourism boards	“Walk there in the shade, or in the sun” guest concierge	Per hotel per month pricing hypothesis: ~$99 per property
Employers of outdoor workers	Shift planner that respects the midday break	Per site per month pricing hypothesis: ~$49
Cities and city planners B2G planning analytics	Where to plant trees or add shade, from the same engine	City licence pricing hypothesis: ~$15,000 per year
Schools, elderly care, events, real estate	Shaded routes and outdoor-time planning; shade and sun reports for sites and venues	Per site subscription
Corporate well-being and insurers	Daylight-walk programmes for staff	Per employee per year pricing hypothesis: ~$12

Prices shown are our pricing hypotheses, not validated quotes. Consumer premium: ~$2.99 per month, no ads.

Freemium. No ads.

No ads, because selling attention would mean selling location.

Keeping location out of any ad pipeline keeps the privacy story clean.

01 Free app for individuals: reach and trust.

02 Optional premium: saved routes and day plans.

03 B2B and B2G subscriptions (fleets, hotels, employers, cities) as the main revenue.

04 A pay-per-route API for delivery and mobility apps.

Cheap to run, cheap to add a city.

SERVER

$6.93

per month: Contabo VPS, 4 vCPU, 8 GB. The VM is shared with other apps, so Zill's share is a fraction. Total fixed cost about $17 per month with the agent.

AI AGENT

$10

per month flat on OpenCode Go: about 3,630 conversations on glm-5.3-flash, about 56 on the strongest model, kimi-k3.

ADD A CITY

$0

Open data is free. Measured: Dubai data bake 7.5 s, shade bake 487.5 s for four dates. Oslo took about 29 minutes, mostly waiting on Overpass.

PER 1,000 USERS

≈ $0.002

of server per 1,000 users: plan price × 0.0003, assuming 20 requests per visit.

Source: docs/COSTS.md. Token counts use characters ÷ 4 on the scripted planner, so output tokens are a lower bound. Bake timings were measured on a dev laptop, not the VM. Per-token API prices are not used (flat subscription). Open-Meteo's free tier is non-commercial, so a paid plan is needed before B2B revenue.

Who needs it, and how many one server holds.

MARKET, SOURCED

70,000+

delivery riders in Dubai (Dubai RTA)

18.72 million

international overnight visitors in 2024 (Dubai DET via WAM)

CAPACITY, MEASURED

WHERE	USERS	REQ/S	P95	ERRORS
Local machine	20	26.4	596 ms	0
Local machine	40	17.1	2.6 s	0
Live VM	10	10.4	1.0 s	0
Live VM	20	16.9	1.6 s	0

One API worker saturates near 26 requests per second on this machine. The live run (capped at 20 concurrent users, 48 s) did not reach its limit, and its latency includes the network round trip. VM: 4 vCPU, 8 GB.

Sources: docs/COSTS.md (scripts/loadtest.py). RTA: rta.ae news, 23 temporary rest areas to protect delivery riders. DET: wam.ae/article/bi46kc8-dubai-welcomes-1872-million-international-visitors. Market figures are context, not a revenue forecast.

NOW · DUBAI AND OSLO

Live prototype at zill.itshamza.me; a pilot with one fleet and one hotel next.

NEXT · UAE AND GULF

Abu Dhabi, Doha, Riyadh; city planners as the first B2G pilot.

THEN · NORTHERN CITIES

A winter daylight pilot in a city like Oslo or Toronto.

Zill & Shams

ظل وشمس

Shade when you need it. Sun when you want it.

ABHAMZAY, TEAM 13: MOHAMMED HAMZA ANSARI · ABHAY PREETHAN

TRY IT: ZILL.ITSHAMZA.ME